YCRY Project Documentation

1 : Project Overview

YCRY is an advanced AI-powered application designed to assist parents in understanding their baby's needs. It leverages Machine Learning to analyze infant cries and interpret them into actionable insights like Hunger, Pain, or Tiredness. The system also features comprehensive child growth tracking, vaccination monitoring, and a generative AI assistant.

2 : Technology Stack

- Backend: Python, Flask
- Frontend: HTML5, CSS3, JavaScript (Vanilla)
- Database: MongoDB Atlas (Cloud NoSQL)
- Machine Learning: TensorFlow/Keras, Librosa (Audio Processing)
- AI Assistant: Google Gemini API
- Visualization: Matplotlib (Spectrograms), Chart.js (Growth Charts)

3 : Backend Architecture (app.py)

The core application logic is handled in 'app.py'. Key components include:

1. Routes:
   - / : Dashboard (Requires Login)
   - /cry : Audio Recording & Analysis Interface
   - /growth : Growth Tracking (Weight/Height) & Charts
   - /vaccine : Vaccination Schedule & Alerts
   - /api/chat : Gemini AI Chat Endpoint

2. Security:
   - Session Management: Flask Sessions
   - Password Hashing: Bcrypt (in database.py)

4 : Database Schema (MongoDB)

Data is stored in the 'users' collection. Each user document contains:

- _id: Email (Unique Identifier)
- password: Hashed String
- caregiver_name: String
- profile: { baby_name, dob, gender, weight_birth, ... }
- growth: [{ date, weight, height }, ...]
- vaccines: [{ name, date }, ...]
- medical_id: { doctor_name, allergies, ... }
- food_log: [{ food, reaction, date }, ...]

5 : AI Model & Preprocessing

1. Input Data Processing (preprocess.py):
   - Audio sampled at 22,050 Hz.
   - Converted to Mel-Spectrograms using Librosa.
   - Augmentation: Noise Injection, Time Stretching.

2. Model Architecture (train_model.py):
 - Type: Convolutional Neural Network (CNN)
 - Input Shape: (64, 64, 3)
 - Layers:
   * 4x Conv2D Blocks (32, 64, 128, 256 filters) with ReLU
   * BatchNormalization & MaxPooling2D after each Conv
   * Dense Layer (256 units) + Dropout (0.5)
   * Output: Softmax (5 Classes: Burping, Discomfort, Hunger, Pain, Tired)

Model Definition Snippet:

```
model = Sequential([
    Input(shape=(64, 64, 3)),
    Conv2D(32, (3, 3), activation='relu'),
    BatchNormalization(),
    MaxPooling2D(2, 2),
    ...
    Dense(5, activation='softmax')
])
```

6 : User Interface

- Dashboard (home.html): Overview of baby's age, next vaccine, and health summary.
- Cry Analysis (cry.html): Audio recorder interface. Displays prediction and advice.
- Growth (growth.html): Interactive charts comparing baby's growth against standard milestones.
- Nutrition (nutrition.html): Age-appropriate food guides and recipe generator.